

Here are the answers to the SQL interview questions:

1. Explain the order of execution of SQL.

SQL queries are executed in the following order:

1. **FROM:** Specifies the tables to query.
2. **WHERE:** Filters the records.
3. **GROUP BY:** Groups the result set based on a condition.
4. **HAVING:** Filters the grouped records.
5. **SELECT:** Selects the columns to display.
6. **DISTINCT:** Removes duplicate records.
7. **ORDER BY:** Sorts the result set.
8. **LIMIT/OFFSET:** Limits the number of records returned.

2. What is the difference between WHERE and HAVING?

- **WHERE** filters rows before any grouping occurs.
- **HAVING** filters rows after grouping (used with aggregate functions).

3. What is the use of GROUP BY?

The **GROUP BY** clause groups rows that have the same values in specified columns into summary rows, like "Total," "Average," etc., often used with aggregate functions like COUNT, SUM, AVG, etc.

4. Explain all types of joins in SQL.

- **INNER JOIN:** Returns records that have matching values in both tables.
- **LEFT JOIN (OUTER JOIN):** Returns all records from the left table and the matching records from the right table. Non-matching rows will contain NULL values.
- **RIGHT JOIN (OUTER JOIN):** Returns all records from the right table and the matching records from the left table. Non-matching rows will contain NULL values.
- **FULL JOIN (OUTER JOIN):** Returns records when there is a match in either left or right table. Non-matching rows will contain NULL values.
- **CROSS JOIN:** Returns the Cartesian product of both tables, combining each row from the first table with each row from the second table.
- **SELF JOIN:** Joins a table with itself.

5. What are triggers in SQL?

A **trigger** is a stored procedure that automatically executes (or fires) when a specified event (INSERT, UPDATE, DELETE) occurs on a table or view.

6. What is a stored procedure in SQL?

A **stored procedure** is a set of SQL statements that can be executed as a single unit. It allows code reusability, simplifies complex queries, and provides security.

7. Explain all types of window functions.

- **ROW_NUMBER():** Assigns a unique sequential integer to rows within a partition of a result set.
- **RANK():** Assigns a rank to each row within a partition, with gaps between ranks for ties.
- **DENSE_RANK():** Similar to RANK() but without gaps in ranks.
- **LEAD():** Accesses the next row in the result set.
- **LAG():** Accesses the previous row in the result set.

8. What is the difference between DROP vs DELETE vs TRUNCATE?

- **DROP:** Removes a table, view, or database entirely, including its data and structure.
- **DELETE:** Removes rows from a table but does not remove the table structure. It can be rolled back.
- **TRUNCATE:** Removes all rows from a table but does not log individual row deletions and cannot be rolled back in most cases.

9. What is the difference between DML, DDL, and DCL?

- **DML (Data Manipulation Language):** Includes SQL commands that modify data (INSERT, UPDATE, DELETE).
- **DDL (Data Definition Language):** Defines database structures (CREATE, ALTER, DROP).
- **DCL (Data Control Language):** Controls access to data (GRANT, REVOKE).

10. Which is faster between CTE and Subquery?

- **CTE (Common Table Expression)** is often more readable and reusable, but the performance difference is negligible in most cases. CTEs are sometimes faster when they are referenced multiple times within the query.
- **Subqueries** might be faster in simple cases where a single value is needed but can be slower when used repeatedly in complex queries.

11. What are constraints and types of constraints?

- **Constraints** are rules enforced on data in a table.
- Types:
 - **PRIMARY KEY:** Uniquely identifies each record.
 - **FOREIGN KEY:** Ensures referential integrity by linking two tables.
 - **UNIQUE:** Ensures that all values in a column are unique.
 - **NOT NULL:** Ensures that a column cannot have NULL values.
 - **CHECK:** Ensures that the value in a column meets a specific condition.

- **DEFAULT:** Provides a default value for a column if no value is specified.

12. What is normalization?

Normalization is the process of organizing data in a database to reduce redundancy and dependency by dividing a database into tables and ensuring relationships between the tables.

13. Difference between Group By and Where Clause?

- **GROUP BY** is used to group rows based on specified columns.
- **WHERE** filters records before grouping, while **HAVING** filters records after grouping.

14. Explain View concepts?

A **view** is a virtual table that consists of a SELECT query. It does not store data but retrieves it dynamically when queried. Views provide a way to simplify complex queries, enhance security, and ensure data consistency.

15. What are different types of constraints?

Refer to question #11 for the types of constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT).

16. Difference between CHAR and VARCHAR?

- **CHAR:** Fixed-length data type, pads extra spaces to the defined length.
- **VARCHAR:** Variable-length data type, does not pad extra spaces.

17. What is an index? Explain its different types.

An **index** improves the speed of data retrieval operations on a database. Types include:

- **Single-column index:** Index on one column.
- **Composite index:** Index on multiple columns.
- **Unique index:** Ensures the uniqueness of the indexed column.
- **Clustered index:** Data rows are stored in the same order as the index.
- **Non-clustered index:** Data rows are stored separately, and the index contains pointers to the data rows.

18. Write an SQL query to find the names of employees starting with 'A'.

```
SELECT name FROM employees WHERE name LIKE 'A%';
```

19. How many types of clauses in SQL?

Common SQL clauses include: **SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, LIMIT, JOIN, ON.**

26. Difference between UNION and UNION ALL in SQL?

- **UNION:** Combines results of two queries and removes duplicates.
- **UNION ALL:** Combines results of two queries without removing duplicates.

27. What are the various types of relationships in SQL?

Types of relationships:

- **One-to-One:** A record in one table is associated with one record in another table.
- **One-to-Many:** A record in one table is associated with multiple records in another table.
- **Many-to-Many:** Records in one table are associated with multiple records in another table.

28. Difference between Primary Key and Foreign Key?

- **Primary Key:** Uniquely identifies each row in a table.
- **Foreign Key:** A column in one table that refers to the primary key of another table.

29. What is the difference between WHERE and HAVING?

Covered in question #2.

30. Find the second highest salary of an employee.

```
SELECT MAX(salary) AS second_highest_salary
```

```
FROM employees
```

```
WHERE salary < (SELECT MAX(salary) FROM employees);
```

31. Difference between Function and Stored Procedure?

- **Function:** Returns a value, can be used in SELECT statements, and cannot modify database state.
- **Stored Procedure:** Does not return a value directly, can perform operations like modifying data, and can contain complex logic.

32. How would you optimize a slow SQL query?

- Use **indexes**.
- Avoid ****SELECT ****.
- Use **EXPLAIN** to analyze the query execution plan.
- Optimize joins and subqueries.
- Ensure proper **data types** for columns.

33. Difference between INNER JOIN and OUTER JOIN?

- **INNER JOIN** returns only matching rows.
- **OUTER JOIN** returns matching rows and the remaining rows with NULL values for non-matching rows.

34. How do you handle duplicate rows in a SQL query?

Use the **DISTINCT** keyword to eliminate duplicate rows:

SELECT DISTINCT column_name FROM table_name;

35. Write a SQL query to find the top 3 departments with the highest average salary.

SELECT department, AVG(salary) AS avg_salary

FROM employees

GROUP BY department

ORDER BY avg_salary DESC

LIMIT 3;

36. Write a SQL query to find the employees who have the same name and work in the same department.

SELECT e1.name

FROM employees e1, employees e2

WHERE e1.name = e2.name AND e1.department = e2.department AND e1.id != e2.id;

37. Write a SQL query to find the departments with no employees.

SELECT department

FROM departments

LEFT JOIN employees ON departments.id = employees.department_id

WHERE employees.id IS NULL;

38. How do you use indexing to improve SQL query performance?

Indexes improve query performance by allowing faster retrieval of rows from a table. Use indexes on columns frequently used in WHERE, JOIN, or ORDER BY clauses.

39. Write a SQL query to find the employees who have worked for more than 5 years.

SELECT name

FROM employees

WHERE hire_date <= CURDATE() - INTERVAL 5 YEAR;

40. What is the difference between SUBQUERY and JOIN?

- **SUBQUERY:** A query inside another query, often used in WHERE, SELECT, or FROM clauses.
- **JOIN:** Combines columns from two or more tables into a single result set.

41. Write a SQL query to find the top 2 products with the highest sales.

SELECT product_name, SUM(sales) AS total_sales

FROM products

GROUP BY product_name

ORDER BY total_sales DESC

LIMIT 2;

42. How do you use stored procedures to improve SQL query performance?

Stored procedures can encapsulate complex logic, reduce network traffic, and improve performance by minimizing repeated queries and allowing efficient batch processing.

43. Write a SQL query to find the customers who have placed an order but have not made a payment.

SELECT customer_id

FROM orders

WHERE order_id NOT IN (SELECT order_id FROM payments);

44. Write a SQL query to find the employees who work in the same department as their manager.

SELECT e.name

FROM employees e

JOIN employees m ON e.department = m.department

WHERE e.manager_id = m.id;

45. How do you use window functions to solve complex queries?

Window functions can help by performing calculations across a set of table rows that are related to the current row, without collapsing the rows. They are useful for running totals, ranking, or calculating moving averages.

46. Write a SQL query to find the top 3 products with the highest average price.

SELECT product_name, AVG(price) AS avg_price

FROM products

GROUP BY product_name

ORDER BY avg_price DESC

LIMIT 3;

47. Write a SQL query to find the employees who have not taken any leave in the last 6 months.

SELECT name

FROM employees

WHERE employee_id NOT IN (SELECT employee_id FROM leaves WHERE leave_date >= CURDATE() - INTERVAL 6 MONTH);

48. Write down various types of relationships in SQL?

Covered in question #27.

49. What is the difference between Cluster and Non-Cluster Index?

- **Clustered Index:** The data rows are sorted in the order of the index.
- **Non-clustered Index:** The data rows are not sorted, and the index contains pointers to the data rows.

50. Why do we use Commit and Rollback commands?

- **COMMIT** saves all changes made during the current transaction.
- **ROLLBACK** undoes changes made during the current transaction.

51. What are indexes in SQL?

Indexes are database objects that improve the speed of data retrieval operations on a table by providing quick access paths to data.

52. What are aggregate functions? Can you name a few?

Aggregate functions perform calculations on multiple rows and return a single value:

- **COUNT(), SUM(), AVG(), MAX(), MIN().**